

# Swapnil Mandekar

Aspiring AI/ML Engineer

Pune, Maharashtra | swapnilavinashmandekar@gmail.com | +91-9604599027



## Professional Summary

Computer Science (Artificial Intelligence & Machine Learning) student with hands-on experience in AI/ML, computer vision, full-stack development and intelligent systems. Built AI-powered applications using Python, React, Flask, SQL, OpenCV, and deep learning, with a strong foundation in Data Structures & Algorithms. Passionate about Generative AI, scalable software systems, and developing technology-driven solutions to real-world problems.

## Education

### Vishwakarma Institute of Technology Pune

Bachelor of Technology in Computer Science (AIML); CGPA: 9.18

Pune, India

2024 -2028

### Vidyardham College

HSC, Class 12th Science: Grade: 77.83%

Maharashtra, India

2021-2023

### Deogiri Global Academy

SSC, Class 10th: Grade: 79.67%

Maharashtra, India

2021

## Experience

### IEEE Computational Intelligence Society (CIS) Pune Section – Internship

June 2026 – July 2026

Developing an AI-powered Answer Sheet Evaluation System that automates the assessment of descriptive answers using Natural Language Processing and Machine Learning techniques. The project focuses on evaluating the quality of the answers through semantic similarity analysis, generating automated scores and feedback, and improving the efficiency and consistency of educational assessment systems.

## Projects

### PlantGuard AI – Plant Disease Detection & Smart Farming Assistant | React, Vite, Flask, Python, CNN, ResNet18, MobileNetV2

- Developed an AI-powered web application for plant disease detection using deep learning models.
- Implemented disease diagnosis, soil analysis, irrigation guidance, multilingual voice assistance, and PDF report generation.
- Designed a responsive React interface using Tailwind CSS for farmers and agricultural users.

### TatvaAI – Multimodal News & Video Analysis Platform | Python, Flask, Node.js, NLP, Deep Learning

- Developed an AI-powered platform for summarizing news articles and videos using multimodal deep learning techniques.
- Integrated speech-to-text, NLP summarization, and transformer-based fake news detection.
- Built scalable backend services using Flask and Node.js with multilingual support.

### Resource Monitor System | Java, JavaFX, JNI, C, Windows APIs

- Built a desktop dashboard for real-time CPU, memory, disk, and process monitoring.
- Integrated Windows APIs using JNI to retrieve low-level system metrics.
- Implemented AI-driven resource monitoring and prediction features.

### Skill Swap – Student Collaboration & Skill Exchange Platform | Flask, MySQL, HTML, CSS

- Built a full-stack platform enabling students to showcase skills and find collaborators.
- Implemented secure authentication, profile management, messaging, and collaboration requests.
- Used advanced SQL queries for user search and collaboration analytics.

## Technical Skills

**Languages:** C, C++, Java, Python, SQL, JavaScript

**AI/ML:** Machine Learning, Deep Learning, Computer Vision, Multimodal AI, CNN, ResNet18, MobileNetV2, Natural Language Processing (NLP), Generative AI, Transfer Learning

**Data:** NumPy, Pandas, OpenCV, Data Preprocessing, Feature Engineering, Data Visualization

**Full-Stack:** React.js, Vite, Flask, Node.js, HTML, CSS, REST APIs, Firebase, MySQL

**Deployment & Tools:** Git, GitHub, Postman, Firebase Authentication

**Core Concepts:** Data Structures & Algorithms, Object-Oriented Programming, DBMS, Operating Systems, Computer Networks

## Achievements & Activities

- Participated in **Google AI Agentic Hackathon**.
- Practicing DSA on **LeetCode** and participated in **GFG** coding events.
- Earned participation certificates in AI/ML and development domains.
- Member of **Ekasutram** (Mathematics Club at VIT), actively participated in club activities and technical discussions.
- Certificates: [View All Certificates](#)